

Homework 6

Solution

1. Compute the value of each of the following series, or prove that it diverges.

(a)

$$\sum_{n=0}^{\infty} (-1)^n \cdot \frac{2^{2n+1} + (-1)^n}{5^n}$$

Solution:

We have

$$\begin{aligned} \sum_{n=0}^{\infty} (-1)^n \cdot \frac{2^{2n+1} + (-1)^n}{5^n} &= \sum_{n=0}^{\infty} (-1)^n \cdot 2 \cdot \frac{4^n}{5^n} + \sum_{n=0}^{\infty} (-1)^n \cdot \frac{(-1)^n}{5^n} \\ &= 2 \cdot \sum_{n=0}^{\infty} \left(-\frac{4}{5}\right)^n + \sum_{n=0}^{\infty} \left(\frac{1}{5}\right)^n = 2 \cdot \frac{1}{1 + \frac{4}{5}} + \frac{1}{1 - \frac{1}{5}} = \frac{10}{9} + \frac{5}{4} = \frac{85}{36} \end{aligned}$$

(b)

$$\sum_{n=1}^{\infty} \cosh\left(\frac{1}{n}\right), \text{ where } \cosh(x) = \frac{e^x + e^{-x}}{2}, \forall x \in \mathbb{R}$$

Solution:

Note that

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \cosh\left(\frac{1}{n}\right) \stackrel{\text{Heine's lemma}}{=} \lim_{x \rightarrow 0} \cosh(x) = \lim_{x \rightarrow 0} \frac{e^x + e^{-x}}{2} = 1$$

Thus, the series diverges by the necessary criterion.

(c)

$$\sum_{n=1}^{\infty} \frac{n^2 - 1}{n^3}$$

Solution:

Note that the series is non-negative.

We have

$$L = \lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{\left\lfloor \frac{n^2 - 1}{n^3} \right\rfloor}{\left\lfloor \frac{1}{n} \right\rfloor} = \lim_{n \rightarrow \infty} \frac{n^3 - n}{n^3} = \lim_{n \rightarrow \infty} \left[1 - \frac{1}{n^2} \right] = 1$$

It follows that the series is divergent together with the Harmonic series.

Alternative solution 1:

We have

$$\begin{aligned}\sum_{n=1}^{\infty} \frac{n^2-1}{n^3} &\geq \sum_{n=1}^{\infty} \frac{n^2-n}{n^3} = \sum_{n=1}^{\infty} \frac{n-1}{n^2} = \sum_{n=0}^{\infty} \frac{n}{(n+1)^2} = \sum_{n=1}^{\infty} \frac{n}{(n+1)^2} \geq \sum_{n=1}^{\infty} \frac{n}{(n+n)^2} \\ &= \sum_{n=1}^{\infty} \frac{n}{4n^2} = \frac{1}{4} \sum_{n=1}^{\infty} \frac{1}{n} = \infty\end{aligned}$$

By the comparison test, $\sum_{n=1}^{\infty} \frac{n^2-1}{n^3} = \infty$.

Alternative solution 2:

We have

$$\sum_{n=1}^{\infty} \frac{n^2-1}{n^3} = \sum_{n=1}^{\infty} \left(\frac{1}{n} - \frac{1}{n^3} \right) = \text{divergent} - \text{convergent} = \text{divergent}$$

(d)

$$\sum_{n=1}^{\infty} \ln \left(\frac{2n+3}{2n+1} \right)$$

Solution:

We have

$$\begin{aligned}\sum_{n=1}^{\infty} \ln \left(\frac{2n+3}{2n+1} \right) &\stackrel{\ln\left(\frac{x}{y}\right)=\ln x - \ln y}{=} \sum_{n=1}^{\infty} [\ln(2n+3) - \ln(2n+1)] \stackrel{\text{definition}}{=} \lim_{N \rightarrow \infty} \sum_{n=1}^N \ln \left(\frac{2n+3}{2n+1} \right) \\ &= \lim_{N \rightarrow \infty} \sum_{n=1}^N [\ln(2n+3) - \ln(2n+1)] \\ &= \lim_{N \rightarrow \infty} [\cancel{\ln(5)} - \ln(3) + \cancel{\ln(7)} - \cancel{\ln(5)} + \cdots + \ln(2N+3) - \cancel{\ln(2N+1)}] \\ &= \lim_{N \rightarrow \infty} [\ln(2N+3) - \ln(3)] \stackrel{\text{Heine's lemma}}{=} \lim_{x \rightarrow \infty} [\ln x - \ln 3] = \infty\end{aligned}$$

Hence the series is divergent.

Alternative solution:

Note that for every $n \in \mathbb{N}$ we have $\frac{2n+3}{2n+1} \geq 1$ which implies that $\ln \left(\frac{2n+3}{2n+1} \right) \geq 0$. It follows that the series is non-negative.

We now compute

$$\begin{aligned}\sum_{n=1}^{\infty} \ln \left(\frac{2n+3}{2n+1} \right) &= \sum_{n=1}^{\infty} \ln \left(1 + \frac{2}{2n+1} \right) \stackrel{\ln(1+x) \geq \frac{x}{x+1}}{\geq} \sum_{n=1}^{\infty} \frac{\left(\frac{2}{2n+1} \right)}{\left(1 + \frac{2}{2n+1} \right)} \\ &= \sum_{n=1}^{\infty} \frac{2}{2n+3} \geq \sum_{n=1}^{\infty} \frac{2}{2n+3n} = \frac{2}{5} \sum_{n=1}^{\infty} \frac{1}{n} = \infty\end{aligned}$$

It follows that the series is divergent, and $\sum_{n=1}^{\infty} \ln \left(\frac{2n+3}{2n+1} \right) = \infty$.

(e)

$$\sum_{n=1}^{\infty} \left[\arcsin \left(\frac{1}{\sqrt{n+1}} \right) - \arcsin \left(\frac{1}{\sqrt{n}} \right) \right]$$

Solution:

We have

$$\begin{aligned} \sum_{n=1}^{\infty} \left[\arcsin \left(\frac{1}{\sqrt{n+1}} \right) - \arcsin \left(\frac{1}{\sqrt{n}} \right) \right] &= \lim_{N \rightarrow \infty} \sum_{n=1}^N \left[\arcsin \left(\frac{1}{\sqrt{n+1}} \right) - \arcsin \left(\frac{1}{\sqrt{n}} \right) \right] \\ &= \lim_{N \rightarrow \infty} \left[\cancel{\arcsin \left(\frac{1}{\sqrt{2}} \right)} - \arcsin(1) + \cancel{\arcsin \left(\frac{1}{\sqrt{3}} \right)} - \cancel{\arcsin \left(\frac{1}{\sqrt{2}} \right)} + \cdots + \arcsin \left(\frac{1}{\sqrt{N+1}} \right) - \cancel{\arcsin \left(\frac{1}{\sqrt{N}} \right)} \right] \\ &= \lim_{N \rightarrow \infty} \left[\arcsin \left(\frac{1}{\sqrt{N+1}} \right) - \arcsin(1) \right] = -\arcsin(1) = -\frac{\pi}{2}. \end{aligned}$$

2. Determine whether each of the following series converges or diverges.

(a)

$$\sum_{n=2}^{\infty} \frac{1}{\ln(n)}$$

Solution:

Note that $\frac{1}{\ln(n)} \geq 0$ for every $n \geq 3$. We now write

$$\sum_{n=3}^{\infty} \frac{1}{\ln(n)} \geq \sum_{n=3}^{\infty} \frac{1}{n} \text{ tail of Harmonic } \infty$$

It follows that the series $\sum_{n=3}^{\infty} \frac{1}{\ln(n)}$ is divergent by the comparison test, and therefore the series $\sum_{n=2}^{\infty} \frac{1}{\ln(n)}$ is divergent as it has a divergent tail.

(b)

$$\sum_{n=1}^{\infty} (\sqrt{n+2} - \sqrt{n})^3$$

Solution:

We note that

$$\sum_{n=1}^{\infty} (\sqrt{n+2} - \sqrt{n})^3 = \sum_{n=1}^{\infty} \left(\frac{n+2-n}{\sqrt{n+2} + \sqrt{n}} \right)^3 = \sum_{n=1}^{\infty} \frac{8}{(\sqrt{n+2} + \sqrt{n})^3}$$

We have

$$= \lim_{n \rightarrow \infty} \frac{\left\lfloor \frac{8}{(\sqrt{n+2} + \sqrt{n})^3} \right\rfloor}{\left\lfloor \frac{1}{n^{\frac{3}{2}}} \right\rfloor} = \lim_{n \rightarrow \infty} \frac{8n^{\frac{3}{2}}}{(\sqrt{n+2} + \sqrt{n})^3} = \lim_{n \rightarrow \infty} \frac{8}{\left(\sqrt{1 + \frac{2}{n}} + 1 \right)^3} = 1$$

Thus, the series converges together with the $\frac{3}{2}$ -Harmonic series.

(c)

$$\sum_{n=1}^{\infty} \frac{1}{(n+e)^n - e^n}$$

Solution:

By the comparison test

$$\sum_{n=3}^{\infty} \frac{1}{(n+e)^n - e^n} \leq \sum_{n=3}^{\infty} \frac{1}{(e+e)^n - e^n} = \sum_{n=3}^{\infty} \frac{1}{2^n \cdot e^n - e^n} \leq \sum_{n=3}^{\infty} \frac{1}{2 \cdot e^n - e^n} = \sum_{n=3}^{\infty} \frac{1}{e^n} \text{ tail of geometric } < \infty$$

It follows that $\sum_{n=3}^{\infty} \frac{1}{(n+e)^n - e^n} < \infty$ and therefore $\sum_{n=1}^{\infty} \frac{1}{(n+e)^n - e^n} < \infty$.

(d)

$$\sum_{n=2}^{\infty} \frac{1}{\ln \left(\tan \left(\frac{\pi}{2} - \frac{1}{n} \right) \right)}$$

You can assume that $\pi > 3$.

Hint. First prove that the series is positive and well-defined.

Solution:

First, as $\frac{1}{n} \leq \frac{1}{2}$ for every $2 \leq n \in \mathbb{N}$, then for every $2 \leq n \in \mathbb{N}$ we have

$$\frac{\pi}{2} > \frac{\pi}{2} - \frac{1}{n} \geq \frac{2\pi}{4} - \frac{2}{4} \stackrel{\pi > 3 > 2}{>} \frac{2\pi}{4} - \frac{\pi}{4} = \frac{\pi}{4}$$

As $\tan$ is strictly increasing on $[\frac{\pi}{4}, \frac{\pi}{2})$, then for every $2 \leq n \in \mathbb{N}$

$$\tan \left(\frac{\pi}{2} - \frac{1}{n} \right) > \tan \left(\frac{\pi}{4} \right) = 1$$

As $\ln$ is increasing, then for every $2 \leq n \in \mathbb{N}$

$$\ln \left(\tan \left(\frac{\pi}{2} - \frac{1}{n} \right) \right) > \ln(1) = 0$$

and therefore the series is non-negative and well-defined.

We note that

$$\lim_{x \rightarrow \infty} \ln \left(\tan \left(\frac{\pi}{2} - \frac{1}{x} \right) \right) \stackrel{t = \tan(\frac{\pi}{2} - \frac{1}{x})}{=} \lim_{t \rightarrow \infty} \ln t = \infty$$

We have

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{\left(\frac{1}{\ln \left(\tan \left(\frac{\pi}{2} - \frac{1}{n} \right) \right)} \right)}{\left(\frac{1}{\ln n} \right)} &= \lim_{n \rightarrow \infty} \frac{\ln n}{\ln \left(\tan \left(\frac{\pi}{2} - \frac{1}{n} \right) \right)} \stackrel{\text{Heine's lemma, } x_n = n}{=} \lim_{x \rightarrow \infty} \frac{\ln x}{\ln \left(\tan \left(\frac{\pi}{2} - \frac{1}{x} \right) \right)} \\ &\stackrel{\frac{?}{L}}{=} \lim_{x \rightarrow \infty} \frac{\left(\frac{1}{x} \right)}{\left(\frac{1}{\tan \left(\frac{\pi}{2} - \frac{1}{x} \right)} \right) \cdot \left(\frac{1}{\cos^2 \left(\frac{\pi}{2} - \frac{1}{x} \right)} \right) \cdot \left(\frac{1}{x^2} \right)} = \lim_{x \rightarrow \infty} \frac{x \tan \left(\frac{\pi}{2} - \frac{1}{x} \right)}{\left(\frac{1}{\cos^2 \left(\frac{\pi}{2} - \frac{1}{x} \right)} \right)} \\ &= \lim_{x \rightarrow \infty} \left[x \tan \left(\frac{\pi}{2} - \frac{1}{x} \right) \cos^2 \left(\frac{\pi}{2} - \frac{1}{x} \right) \right] = \lim_{x \rightarrow \infty} \left[x \cdot \frac{\sin \left(\frac{\pi}{2} - \frac{1}{x} \right)}{\cos \left(\frac{\pi}{2} - \frac{1}{x} \right)} \cdot \cos^2 \left(\frac{\pi}{2} - \frac{1}{x} \right) \right] \\ &= \lim_{x \rightarrow \infty} \left[x \sin \left(\frac{\pi}{2} - \frac{1}{x} \right) \cos \left(\frac{\pi}{2} - \frac{1}{x} \right) \right] = \lim_{x \rightarrow \infty} \left[x \cos \left(\frac{1}{x} \right) \sin \left(\frac{1}{x} \right) \right] \\ &= \lim_{x \rightarrow \infty} \cos \left(\frac{1}{x} \right) \cdot \frac{\sin \left(\frac{1}{x} \right)}{\left(\frac{1}{x} \right)} \stackrel{t = \frac{1}{x}}{=} \lim_{t \rightarrow 0^+} \left[\cos t \cdot \frac{\sin t}{t} \right] = 1 \cdot 1 = 1. \end{aligned}$$

It follows that the series diverges together with the series $\sum_{n=2}^{\infty} \frac{1}{\ln(n)}$.

Alternative solution (after showing that the series is non-negative and well-defined):

As $x \cos x \leq \sin x$ for every $0 < x < \frac{\pi}{2}$, then $\frac{\cos x}{\sin x} \leq \frac{1}{x}$ for every $0 < x < \frac{\pi}{2}$.

As, for every $2 \leq n \in \mathbb{N}$ we have $0 < \frac{1}{n} \leq \frac{1}{2} <^{\pi > 3 > 1} \frac{\pi}{2}$, then for every $2 \leq n \in \mathbb{N}$ we have

$$\tan\left(\frac{\pi}{2} - \frac{1}{n}\right) = \cot\left(\frac{1}{n}\right) = \frac{\cos\left(\frac{1}{n}\right)}{\sin\left(\frac{1}{n}\right)} \leq \frac{1}{\left(\frac{1}{n}\right)} = n$$

As $\ln$ is strictly increasing, then for every $2 \leq n \in \mathbb{N}$

$$\ln\left(\tan\left(\frac{\pi}{2} - \frac{1}{n}\right)\right) \leq \ln n$$

and therefore (both sides are positive)

$$\frac{1}{\ln\left(\tan\left(\frac{\pi}{2} - \frac{1}{n}\right)\right)} \geq \frac{1}{\ln n}$$

thus

$$\sum_{n=2}^{\infty} \frac{1}{\ln\left(\tan\left(\frac{\pi}{2} - \frac{1}{n}\right)\right)} \geq \sum_{n=2}^{\infty} \frac{1}{\ln n} = \infty$$

Thus the series $\sum_{n=2}^{\infty} \frac{1}{\ln\left(\tan\left(\frac{\pi}{2} - \frac{1}{n}\right)\right)}$ diverges.

(e)

$$\sum_{n=1}^{\infty} \int_0^{\frac{1}{n}} \frac{\sin^2(t)}{t} dt$$

First prove that the sequence is well-defined.

Solution:

First, as $\lim_{t \rightarrow 0} \frac{\sin^2(t)}{t} \stackrel{0}{=} \lim_{t \rightarrow 0} \frac{2 \sin(t) \cos(t)}{1} = 0$, then the function defined by $f(t) = \frac{\sin^2(t)}{t}$ for every $t > 0$ has a removable discontinuity at $t = 0$. Thus, the integral $\int_0^{\frac{1}{n}} \frac{\sin^2(t)}{t} dt$ is well-defined for every $n \in \mathbb{N}$.

As $\sin(t) \leq t$ for every $t > 0$, then $\frac{\sin^2(t)}{t} \leq t$ for every $t > 0$. By the monotonicity of the definite integral (the point $t = 0$ doesn't affect the integral), we have

$$0 \leq \int_0^{\frac{1}{n}} \frac{\sin^2(t)}{t} dt \leq \int_0^{\frac{1}{n}} t dt = \left[\frac{t^2}{2} \right]_0^{\frac{1}{n}} = \frac{1}{2n^2}, \quad \forall n \in \mathbb{N}$$

Thus, by the comparison principle we have

$$\sum_{n=1}^{\infty} \int_0^{\frac{1}{n}} \frac{\sin^2(t)}{t} dt \leq \sum_{n=1}^{\infty} \frac{1}{2n^2} = \frac{1}{2} \sum_{n=1}^{\infty} \frac{1}{n^2} < \infty$$

3. (a) Student A has proved that $\int_1^\infty \frac{dx}{x} = 0$ in the following way:

$$\int_1^\infty \frac{dx}{x} = \int_1^2 \frac{dx}{x} + \int_2^3 \frac{dx}{x} + \int_3^4 \frac{dx}{x} + \dots = \ln(2) - \ln(1) + \ln(3) - \ln(2) + \ln(4) - \ln(3) + \dots = 0$$

Explain student A's mistake.

Solution:

Student A made the following errors in his proof:

- i. Student A did not explain why $\int_1^\infty \frac{dx}{x} = \int_1^2 \frac{dx}{x} + \int_2^3 \frac{dx}{x} + \int_3^4 \frac{dx}{x} + \dots$

This equality assumes that $\int_1^\infty \frac{dx}{x}$ converges in the extended sense, which renders the proof circular.

- ii. The equality

$$\int_1^\infty \frac{dx}{x} = \int_1^2 \frac{dx}{x} + \int_2^3 \frac{dx}{x} + \int_3^4 \frac{dx}{x} + \dots = \ln(2) - \ln(1) + \ln(3) - \ln(2) + \ln(4) - \ln(3) + \dots$$

is a mistake, as he should have added parenthesis on every two terms, namely,

$$\int_1^2 \frac{dx}{x} + \int_2^3 \frac{dx}{x} + \int_3^4 \frac{dx}{x} + \dots = (\ln 2 - \ln 1) + (\ln 3 - \ln 2) + (\ln 4 - \ln 3) + \dots$$

- iii. Student A ignored the definition of an infinite sum as a limit of finite sums. What he should have is

$$(\ln 2 - \ln 1) + (\ln 3 - \ln 2) + (\ln 4 - \ln 3) + \dots = \sum_{n=1}^{\infty} (\ln(n+1) - \ln n) =$$

$$\lim_{N \rightarrow \infty} \sum_{n=1}^N (\ln(n+1) - \ln n) = \lim_{N \rightarrow \infty} (\ln(N+1) - \ln 1) = \infty$$

- (b) Student A has proved that $\sum_{n=0}^{\infty} \frac{1}{2^n} = 2$ in the following way:

Denote $s = \sum_{n=0}^{\infty} \frac{1}{2^n} = 1 + \frac{1}{2} + \frac{1}{4} + \dots$. Then $s - 1 = \frac{1}{2} + \frac{1}{4} + \dots$. Thus, $2(s - 1) = 1 + \frac{1}{2} + \dots = s$.

It follows that $2s - 2 = s$. Thus $s = 2$.

Inspired by his method, student A also proved that $\sum_{n=0}^{\infty} 2^n = -1$ in the following way:

Denote $s = \sum_{n=0}^{\infty} 2^n = 1 + 2 + 4 + \dots$. Then $s - 1 = 2 + 4 + 8 + \dots$. Thus, $\frac{1}{2}(s - 1) = 1 + 2 + 4 + \dots = s$.

It follows that $\frac{1}{2}(s - 1) = s$. Thus $s = -1$.

Find the mistake of student A.

Solution:

Student A assumed that the series is convergent, but while this assertion is true for the first series (geometric with $q = \frac{1}{2}$) it is false for the second series (indeed, the necessary criterion does not hold).

4. Let $(a_n)_{n=1}^{\infty}$ be a sequence of real numbers. Consider the sequence defined by

$$b_n = \begin{cases} a_k & n = 2k - 1 \\ -a_k & n = 2k \end{cases} \quad \forall n \in \mathbb{N}$$

Prove that the series $\sum_{n=1}^{\infty} b_n$ is convergent if and only if $\lim_{n \rightarrow \infty} b_n = 0$.

Solution:

$\Rightarrow$: As $\sum_{n=1}^{\infty} b_n$ is convergent then $\lim_{n \rightarrow \infty} b_n = 0$ by the necessary criterion.

$\Leftarrow$: As $\lim_{n \rightarrow \infty} b_n = 0$, then by the inheritance theorem we have $\lim_{k \rightarrow \infty} a_k = 0$.

Consider the sequence defined by

$$S_N = \sum_{n=1}^N b_n = b_1 + b_2 + \dots + b_N, \quad \forall N \in \mathbb{N}$$

Note that

$$S_{2N} = a_1 - a_1 + a_2 - a_2 + \dots + a_N - a_N = 0, \quad \forall N \in \mathbb{N}$$

and therefore $\lim_{N \rightarrow \infty} S_{2N} = 0$.

Note also that

$$S_{2N} = S_{2N-1} + b_{2N} = S_{2N-1} - a_N \Rightarrow S_{2N-1} = S_{2N} + a_N, \quad \forall N \in \mathbb{N}$$

and therefore $\lim_{N \rightarrow \infty} S_{2N-1} = \lim_{N \rightarrow \infty} [S_{2N} + a_N] = 0 + 0 = 0$. It follows that $\lim_{N \rightarrow \infty} S_N = 0$ and therefore

$$\sum_{n=1}^{\infty} b_n = \lim_{N \rightarrow \infty} S_N = 0$$

5. Let $(a_n)_{n=1}^{\infty}$ be a sequence of real numbers such that $\lim_{n \rightarrow \infty} a_n = 0$.

Let $a, b, c \in \mathbb{R}$ such that $a + b + c = 0$.

Let $(b_n)_{n=1}^{\infty}$ be a sequence of real numbers such that for every $n \in \mathbb{N}$ we have $b_n = a \cdot a_n + b \cdot a_{n+1} + c \cdot a_{n+2}$.

Prove that the series $\sum_{n=1}^{\infty} b_n$ converges and find its value.

Solution:

As $a + b + c = 0$, then $c = -a - b$, thus for every $n \in \mathbb{N}$ we have

$$b_n = a \cdot a_n + b \cdot a_{n+1} + (-a - b) \cdot a_{n+2} = a \cdot a_n + b \cdot a_{n+1} - a \cdot a_{n+2} - b a_{n+2} = a (a_n - a_{n+2}) + b (a_{n+1} - a_{n+2})$$

We note that for every $n \in \mathbb{N}$ we have

$$\sum_{n=1}^N b_n = \sum_{n=1}^N [a (a_n - a_{n+2}) + b (a_{n+1} - a_{n+2})] = a \sum_{n=1}^N (a_n - a_{n+2}) + b \sum_{n=1}^N (a_{n+1} - a_{n+2}) = a (a_1 + a_2 - a_{N+1} - a_{N+2}) + b (a_2 - a_{N+2})$$

As $\lim_{n \rightarrow \infty} a_n = 0$, then $\lim_{n \rightarrow \infty} a_{n+1} = \lim_{n \rightarrow \infty} a_{n+2} = 0$ and therefore

$$\sum_{n=1}^{\infty} b_n = \lim_{N \rightarrow \infty} \sum_{n=1}^N b_n = \lim_{N \rightarrow \infty} [a (a_1 + a_2 - a_{N+1} - a_{N+2}) + b (a_2 - a_{N+2})] \stackrel{\text{AOL}}{=} a \cdot (a_1 + a_2) + b \cdot a_2$$

Thus the series converges and its value is $a \cdot (a_1 + a_2) + b \cdot a_2$.

6. Let $(a_n)_{n=1}^{\infty}$ $(b_n)_{n=1}^{\infty}$ be two sequences of real numbers.

Prove or disprove each of the following statements:

(a) Suppose that $(a_n)_{n=1}^{\infty}$ $(b_n)_{n=1}^{\infty}$ are non-negative.

Suppose that $a_n < b_n$ for every $n \in \mathbb{N}$ and $\sum_{n=1}^{\infty} b_n < \infty$.

Then $\sum_{n=1}^{\infty} a_n < \sum_{n=1}^{\infty} b_n$.

Solution:

True. As $a_n < b_n$ for every $n \in \mathbb{N}$, then $a_1 < b_1$.

As $\sum_{n=1}^{\infty} b_n < \infty$, then $\sum_{n=2}^{\infty} b_n < \infty$.

As $a_n < b_n$ for every $n \in \mathbb{N}$ and $\sum_{n=2}^{\infty} b_n < \infty$, then by the comparison principle $\sum_{n=2}^{\infty} a_n < \infty$ and

$$\sum_{n=2}^{\infty} a_n \leq \sum_{n=2}^{\infty} b_n < \infty$$

thus

$$a_1 + \sum_{n=2}^{\infty} a_n < b_1 + \sum_{n=2}^{\infty} b_n$$

$$\sum_{n=1}^{\infty} a_n < \sum_{n=1}^{\infty} b_n$$

(b) Consider the sequence defined by $S_N = \sum_{n=1}^N a_n$ for every $N \in \mathbb{N}$.

Suppose that $b_n = S_n^2 - 2S_{n-1} + 3$ for every $2 \leq n \in \mathbb{N}$.

If the series $\sum_{n=2}^{\infty} b_n$ is convergent, then the series $\sum_{n=1}^{\infty} a_n$ is divergent.

Solution:

True. ATC that the series $\sum_{n=1}^{\infty} a_n$ is convergent. Then $\exists s \in \mathbb{R}$ such that $\lim_{n \rightarrow \infty} S_n = s$.

As the series $\sum_{n=1}^{\infty} b_n$ is convergent, then by the necessary criterion $\lim_{n \rightarrow \infty} b_n = 0$. Thus, by AOL,

$$0 = \lim_{n \rightarrow \infty} b_n = \lim_{n \rightarrow \infty} (S_n^2 - 2S_{n-1} + 3) = s^2 - 2s + 3$$

Contradiction, as the equation $s^2 - 2s + 3 = 0$ has no real solutions.

(c) Suppose that $\sum_{n=1}^{\infty} a_n$ is convergent. Then the series $\sum_{n=1}^{\infty} a_{2n}$ is convergent.

Solution:

False. Consider the sequence defined by

$$a_n = \begin{cases} \frac{1}{k} & n = 2k - 1 \\ -\frac{1}{k} & n = 2k \end{cases} \quad \forall n \in \mathbb{N}$$

Note that

$$\lim_{k \rightarrow \infty} a_{2k-1} = \lim_{k \rightarrow \infty} \frac{1}{k} = 0, \quad \text{and} \quad \lim_{k \rightarrow \infty} a_{2k} = \lim_{k \rightarrow \infty} \left(-\frac{1}{k}\right) = 0$$

It follows that $\lim_{n \rightarrow \infty} a_n = 0$, and therefore, by Q4, the series $\sum_{n=1}^{\infty} a_n$ is convergent. However

$$\sum_{n=1}^{\infty} a_{2n} = \sum_{n=1}^{\infty} \left(-\frac{1}{n}\right) \text{ minus Harmonic } \equiv \text{ diverges}$$

(d) Suppose that $\sum_{n=1}^{\infty} a_n$ is convergent. Then the series $\sum_{n=1}^{\infty} \frac{a_{2n-1} + a_{2n}}{2}$ is convergent.

Solution:

True. First, as the series $\sum_{n=1}^{\infty} a_n$ is convergent then there exists an $s \in \mathbb{R}$ such that $s = \lim_{N \rightarrow \infty} \sum_{n=1}^N a_n$.

Consider the sequence $(S_N)_{N=1}^{\infty}$ defined by

$$S_N = \sum_{n=1}^N a_n, \quad \forall N \in \mathbb{N}.$$

Then $\lim_{N \rightarrow \infty} S_N = s$, and therefore, by the Inheritance theorem, $\lim_{N \rightarrow \infty} S_{2N} = s$. It follows that

$$\begin{aligned} \sum_{n=1}^{\infty} \frac{a_{2n-1} + a_{2n}}{2} &= \lim_{N \rightarrow \infty} \left[\frac{a_1 + a_2}{2} + \frac{a_3 + a_4}{2} + \cdots + \frac{a_{2N-1} + a_{2N}}{2} \right] = \lim_{N \rightarrow \infty} \frac{1}{2} [(a_1 + a_2) + (a_3 + a_4) + \cdots + (a_{2N-1} + a_{2N})] \\ &= \lim_{N \rightarrow \infty} \frac{1}{2} [(a_1 + a_2 + \cdots + a_{2N})] = \lim_{N \rightarrow \infty} \frac{1}{2} S_{2N} \stackrel{\text{AOL}}{=} \frac{1}{2} s. \end{aligned}$$

It follows that the series $\sum_{n=1}^{\infty} \frac{a_{2n-1} + a_{2n}}{2}$ is convergent, and

$$\sum_{n=1}^{\infty} \frac{a_{2n-1} + a_{2n}}{2} = \frac{1}{2} \sum_{n=1}^{\infty} a_n = \frac{1}{2} s.$$